Spectral Proper Orthogonal Decomposition of Wind-over-Waves DNS Data

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Abstract. This report presents a Spectral Proper Orthogonal Decomposition (SPOD) analysis of a Direct Numerical Simulation dataset of a wind-over-waves boundary layer. The decomposition is performed on a $192 \times 192 \times 65$ grid using 500 temporal snapshots, yielding 26 resolved frequencies from $N_{\text{blk}} = 19$ overlapping Welch blocks. The weighted Method of Snapshots is implemented with a non-uniform spatial weight matrix to compensate for the wall-normal grid clustering. The analysis identifies a sharp global energy peak at $f = 2.67$, where the leading mode captures 33.9% of the local spectral energy and corresponds to a streamwise-dominated coherent structure ($U : V : W = 68 : 22 : 10$). A coherent–incoherent transition is identified at $f \approx 18.67$ from the mode-1 coherence ratio. A localised anomaly is found at $f = 16.0$, where the leading mode collapses to a surface-locked structure with the wall-normal component reaching 35.5% — the only frequency in the spectrum where W exceeds V. A systematic anisotropy cascade is observed: the streamwise-dominated coherent regime gives way to a near-isotropic ($U : V : W \approx 27 : 40 : 33$) incoherent plateau at high frequencies, with the streamwise component crossing equipartition at $f = 26.67$. Phase-energy diagnostics confirm that all examined modes correspond to downstream-propagating waves rather than standing structures.

1 Introduction

1.1 Motivation

Standard Proper Orthogonal Decomposition (POD) identifies the most energetic spatial structures of a flow field but ignores temporal evolution; modes that mix multiple physical mechanisms operating at different frequencies are not separated by the decomposition. Spectral POD (SPOD) addresses this limitation by retaining both spatial and temporal information[2, 1]. Applying a temporal Fourier transform to the space–time correlation tensor yields the Cross-Spectral Density (CSD) tensor $\hat{C}(\boldsymbol{x}, \boldsymbol{x}', f)$, whose eigendecomposition produces modes that are coherent in both space and time. SPOD therefore isolates physically meaningful phenomena — such as shedding vortices, traveling waves, or wave-coherent boundary-layer structures — at specific frequencies.

This report applies SPOD to a Direct Numerical Simulation (DNS) dataset of a wind-over-waves boundary layer. The grid contains $N_x \times N_y \times N_z = 192 \times 192 \times 65$ points, and the snapshot ensemble comprises 500 temporal samples covering the streamwise, spanwise, and wall-normal velocity components (U, V, W). The objective is to identify the dominant coherent structures of the boundary layer, quantify their wall-normal distribution, and characterise the transition from coherent to incoherent dynamics across the resolved frequency band.

1.2 Welch's Method for Cross-Spectral Density Estimation

A reliable estimate of the CSD requires an ensemble of statistically independent realisations. Since the DNS produces a single continuous time series, Welch's method is employed to construct this ensemble [4, 1]. The full record is partitioned into N_{blk} overlapping blocks, each of which is independently Fourier-transformed in time. This procedure provides a controlled compromise between frequency resolution and statistical noise reduction: longer blocks improve resolution at the cost of fewer realisations, while more numerous short blocks reduce variance at the cost of resolution. For a selected frequency f_j, the Fourier coefficients across all blocks are gathered into the frequency-domain snapshot matrix

$$X(f_j) = \left[\hat{\boldsymbol{q}}^{(1)}(f_j),\, \hat{\boldsymbol{q}}^{(2)}(f_j),\, \ldots,\, \hat{\boldsymbol{q}}^{(N_{\text{blk}})}(f_j)\right] \in \mathbb{C}^{3N \times N_{\text{blk}}}.$$

1.3 The Spatial Weight Matrix

The weight matrix W defines the inner product under which the spatial modes are optimal and orthogonal. In an incompressible-flow SPOD analysis, W acts as a discrete spatial quadrature operator that accounts for the volume of influence of each grid point. This correction is essential whenever the computational mesh is non-uniform: in the present DNS, the wall-normal direction is clustered near the surface to resolve the boundary-layer shear, so that without spatial weighting the densely-packed near-surface nodes would dominate the kinetic-energy calculation purely due to grid density rather than physical content. The construction of W for the present grid is detailed in Section 4.2.

[1] Source code available at: https://github.com/sudinbhujushrestha/SPOD.machine.learning.project

2 Theoretical Formulation

This section presents the weighted Method of Snapshots, which forms the mathematical backbone of the SPOD pipeline. The derivation tracks how the grid weight matrix W enters both the reduced eigenvalue problem and the final spatial mode reconstruction.

2.1 Weighted Method of Snapshots

The theoretical weighted SPOD eigenvalue problem at a specific frequency f_j is

$$\hat{C}W\hat{\Phi} = \hat{\Phi}\hat{\Lambda}. \tag{1}$$

Substituting the empirical CSD estimate $\hat{C} = \frac{1}{N_{\text{blk}}}XX^*$ yields

$$\frac{1}{N_{\text{blk}}}XX^*W\hat{\Phi} = \hat{\Phi}\hat{\Lambda}. \tag{2}$$

Because XX^*W is an asymmetric matrix of size $3N \times 3N$, this system cannot be solved efficiently by direct decomposition. Multiplying both sides on the left by X^*W gives

$$X^*W\left(\frac{1}{N_{\text{blk}}}XX^*W\hat{\Phi}\right) = X^*W\hat{\Phi}\hat{\Lambda}, \tag{3}$$

and using associativity,

$$\frac{1}{N_{\text{blk}}}\left(X^*WX\right)\left(X^*W\hat{\Phi}\right) = \left(X^*W\hat{\Phi}\right)\hat{\Lambda}. \tag{4}$$

Defining the reduced correlation matrix $R \equiv \frac{1}{N_{\text{blk}}}X^*WX$ and the reduced eigenvectors $\Psi \equiv X^*W\hat{\Phi}$, this becomes

$$R\Psi = \Psi\hat{\Lambda}. \tag{5}$$

The reduced system is computationally tractable because R is only $N_{\text{blk}} \times N_{\text{blk}}$. The physical influence of the grid weights W is embedded inside R. Solving Equation (5) yields the modal energies $\hat{\Lambda}$ and the orthonormal reduced eigenvectors Ψ (satisfying $\Psi^*\Psi = I$).

Spatial mode reconstruction

Because X^*W is rectangular, the definition $\Psi = X^*W\hat{\Phi}$ cannot be inverted directly to recover the spatial modes. Following Sirovich's snapshot-method assumption[8], the spatial modes are constructed as a linear combination of the original frequency snapshots, weighted by the reduced eigenvectors:

$$\hat{\Phi}_{\text{raw}} = X\Psi. \tag{6}$$

Orthonormality scaling

The final spatial modes must satisfy the weighted discrete orthonormality condition $\hat{\Phi}^*W\hat{\Phi} = I$. Testing the raw modes,

$$\hat{\Phi}^*_{\text{raw}}W\hat{\Phi}_{\text{raw}} = (X\Psi)^*W(X\Psi) = \Psi^*(X^*WX)\Psi. \tag{7}$$

From Equation (5), $X^*WX\,\Psi = N_{\text{blk}}\Psi\hat{\Lambda}$, giving

$$\hat{\Phi}^*_{\text{raw}}W\hat{\Phi}_{\text{raw}} = N_{\text{blk}}\Psi^*\Psi\hat{\Lambda} = N_{\text{blk}}\hat{\Lambda}. \tag{8}$$

The squared length of the raw modes is therefore scaled by $N_{\text{blk}}\hat{\Lambda}$. Because the orthonormality condition is quadratic in $\hat{\Phi}$, the correctly scaled SPOD modes are

$$\hat{\Phi} = \frac{1}{\sqrt{N_{\text{blk}}\hat{\Lambda}}}X\Psi. \tag{9}$$

Equation (5) delivers the modal energies $\hat{\Lambda}$, and Equation (9) delivers the spatial modes $\hat{\Phi}$.

2.2 Procedure Summary

The complete SPOD procedure derived above is summarised schematically in Figure 1.

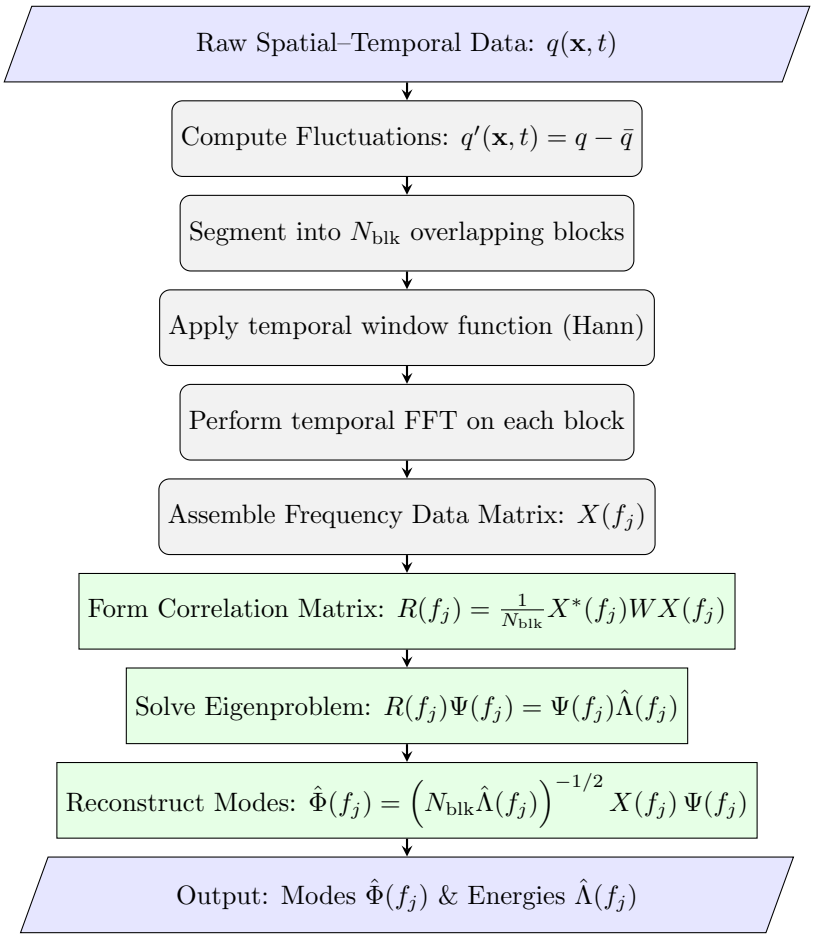

Figure 1: SPOD procedure flowchart. Blue trapezoids denote data inputs and outputs; grey blocks denote preprocessing steps; green blocks denote the core mathematical operations of the weighted Method of Snapshots.

3 Results

This section presents the SPOD analysis of the wind-over-waves DNS dataset. The discussion proceeds from a global characterisation of the eigenvalue spectrum through quantitative diagnostics of mode dominance, to the spatial structure of the leading modes and their component-energy distribution. The pipeline produced 26 frequency bins from $N_{\text{blk}} = 19$ overlapping blocks, with frequency resolution $\Delta f = 2.667$ and Nyquist frequency $f_{\text{Ny}} = 66.67$.

3.1 SPOD Eigenvalue Spectrum

Figure 2 presents the complete SPOD eigenvalue spectrum across all 26 resolved frequencies, displayed on both linear-frequency/log-energy and log–log axes.

The spectrum reveals three distinct frequency regimes. At low frequencies ($f \lesssim 5.33$), the leading mode is well separated from the remaining modes, with the global energy peak at $f = 2.67$ where $\hat{\Lambda}_1 = 8.84 \times 10^{-3}$ and the coherence ratio reaches $\hat{\Lambda}_1 / \sum_k \hat{\Lambda}_k = 0.339$. This separation identifies a coherent structure concentrated at a single frequency. An intermediate coherent regime extends up to $f \approx 16.0$, where the leading mode retains a meaningful energy advantage over the sub-leading modes. Beyond $f \approx 18.67$, the individual mode curves converge and the spectrum transitions into an incoherent turbulent regime in which energy is distributed nearly uniformly across all 19 mode ranks — the spectral signature of broadband, spatially uncorrelated fluctuations.

A localised secondary elevation is clearly visible near $f = 16.0$ in both mode 1 and mode 2, breaking the otherwise monotonic decay of the leading-mode curve. This bump corresponds to the highest rank-separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$ in the spectrum (Section 3.2) and motivates the inclusion of $f = 16.0$ as a target frequency in Section 3.3.

On the log–log axes, the total energy curve $\sum_k \hat{\Lambda}_k$ decays with a slope broadly consistent with the $f^{-5/3}$ Kolmogorov inertial subrange reference from $f \approx 6$ up to the Nyquist frequency $f = 66.67$, indicating that the

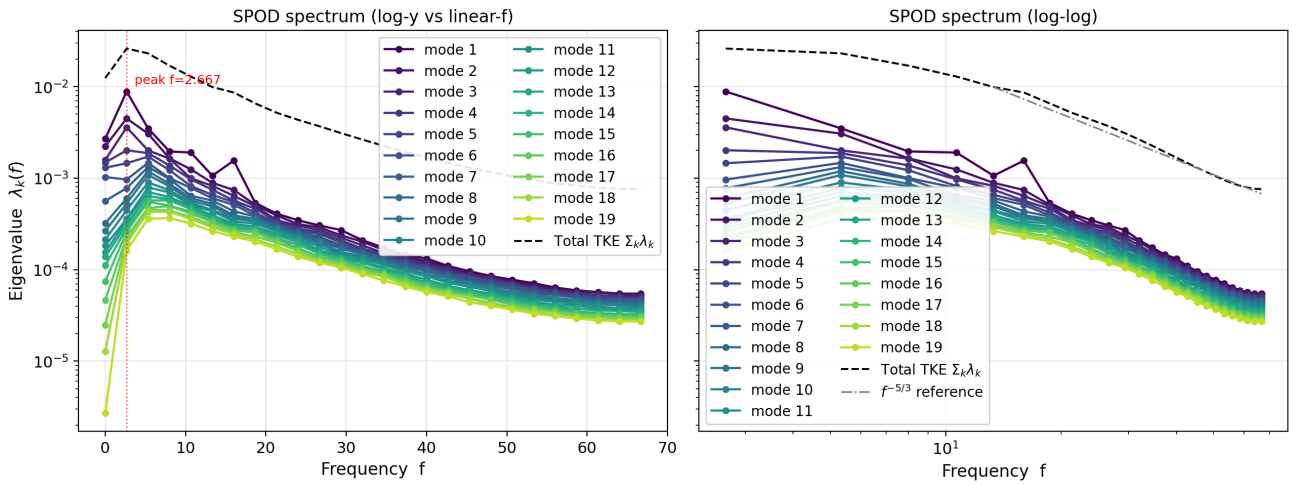

Figure 2: SPOD eigenvalue spectrum across all resolved frequencies. Each curve corresponds to a single mode rank (modes 1–19, darkest to lightest); the dashed black line is the total energy per frequency $\sum_k \hat{\Lambda}_k$. Left: log-energy versus linear frequency, with the global peak at $f = 2.67$ (red dotted line). Right: log–log axes, with a dash-dotted $f^{-5/3}$ reference line.

small-scale fluctuations are well-resolved across the sampled bandwidth.

3.2 Spectral Energy Distribution and Mode Dominance

The modal energy distribution was analysed across the discrete frequency bins using three diagnostic ratios: the leading-mode energy $\hat{\Lambda}_1$, the mode-1 coherence ratio $\hat{\Lambda}_1/\sum_k \hat{\Lambda}_k$, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2$. The total summed modal energy across the spectrum is $\sum_f \sum_k \hat{\Lambda}_k(f) = 1.51 \times 10^{-1}$. As shown in Table 1, the spectrum exhibits a sharp global peak at $f = 2.67$.

Table 1: Top five frequencies ranked by leading mode energy $\hat{\Lambda}_1$. The $f = 0$ entry reflects residual block-to-block variation in the mean velocity field rather than a true zero-frequency mode, since each block undergoes independent mean subtraction prior to the FFT.

Rank	f	$\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1$	$\hat{\Lambda}_1/\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1/\hat{\Lambda}_2$	**Modes** $\geq 90\%$
1	2.6667	2.606e-02	8.835e-03	0.339	1.97	10
2	5.3333	2.309e-02	3.493e-03	0.151	1.14	14
3	0.0000	1.241e-02	2.692e-03	0.217	1.21	8
4	8.0000	1.692e-02	1.958e-03	0.116	1.19	16
5	10.667	1.289e-02	1.902e-03	0.148	1.53	16

At the peak frequency $f = 2.67$, the leading mode captures 33.9% of the local spectral energy, and the rank separation $\hat{\Lambda}_1/\hat{\Lambda}_2 = 1.97$ is just below the ≥ 2.0 threshold conventionally used to define rank-1 dominance. The strongest separation in the entire spectrum occurs at $f = 16.0$, with $\hat{\Lambda}_1/\hat{\Lambda}_2 = 2.09$, suggesting that this frequency hosts a particularly distinct coherent structure despite its lower absolute energy. These two frequencies, together with $f = 8.0$ (the broader coherent regime) and $f = 53.33$ (a turbulent reference), define the four target frequencies analysed in detail in subsequent sections.

The mode-1 coherence ratio identifies a clear coherent–incoherent transition. With $N_{\text{blk}} = 19$ blocks, the incoherent baseline is $1/N_{\text{blk}} = 0.053$ and the conventional coherence threshold is $2/N_{\text{blk}} = 0.105$. The ratio $\hat{\Lambda}_1/\sum_k \hat{\Lambda}_k$ exceeds this threshold only for $f \leq 16.0$, with the first crossing below it occurring at $f = 18.67$. Beyond this frequency, the energy is distributed nearly uniformly across all blocks, and the number of modes required to capture 90% of the variance saturates at 16–17 — close to the maximum of $N_{\text{blk}} = 19$. This is the spectral signature of broadband, spatially uncorrelated turbulent fluctuations and defines the upper edge of the coherent regime at $f \approx 18.67$.

3.3 Vertical Structure of the Three Leading Modes Across Frequencies

The vertical organisation of the three most energetic SPOD modes was examined as a function of frequency. Figures 3, 4, and 5 present the phase-aligned real parts of $\hat{\Phi}_1$, $\hat{\Phi}_2$, and $\hat{\Phi}_3$ at four target frequencies ($f = 2.67$,

8.0, 16.0, and 53.33), sliced at mid-span ($y = 4.737$). The four frequencies sample the three spectral regimes identified in Section 3.1: the primary coherent peak ($f = 2.67$), the intermediate coherent range ($f = 8.0$ and $f = 16.0$), and the incoherent turbulent reference ($f = 53.33$).

Canonical phase alignment

SPOD modes are complex-valued: each grid point has both a magnitude and a phase angle, encoding the amplitude and temporal oscillation of a structure. To compare modes across frequencies fairly, we need to rotate each mode to a canonical phase state.

A useful diagnostic is the real-part energy fraction: if we decompose the mode into its real and imaginary parts separately and calculate what fraction of the total energy is in the real part, this tells us the wave type. A value near 100% means the energy is almost entirely in the real part - this is a standing wave with no temporal oscillation. A value near 50% means the energy is split equally between real and imaginary parts - this is a purely travelling wave that oscillates forward and backward in time as it propagates. An intermediate value like 30% means the mode has both standing and travelling components, with the travelling component dominant.

To visualise each mode at the same canonical phase, we apply a global phase rotation $e^{-i\varphi^\star}$ chosen to maximise the real-part energy:

$$\varphi^\star = \tfrac{1}{2} \arg\left(\sum_n W_n \, \hat{\Phi}_n^2 \right), \tag{10}$$

where W_n are the spatial weights from Section 4.2.

Across all twelve mode–frequency combinations examined, the real-part energy fraction is approximately 50% (within 0.3% of the theoretical value), confirming that every plotted structure is a downstream-propagating wave.

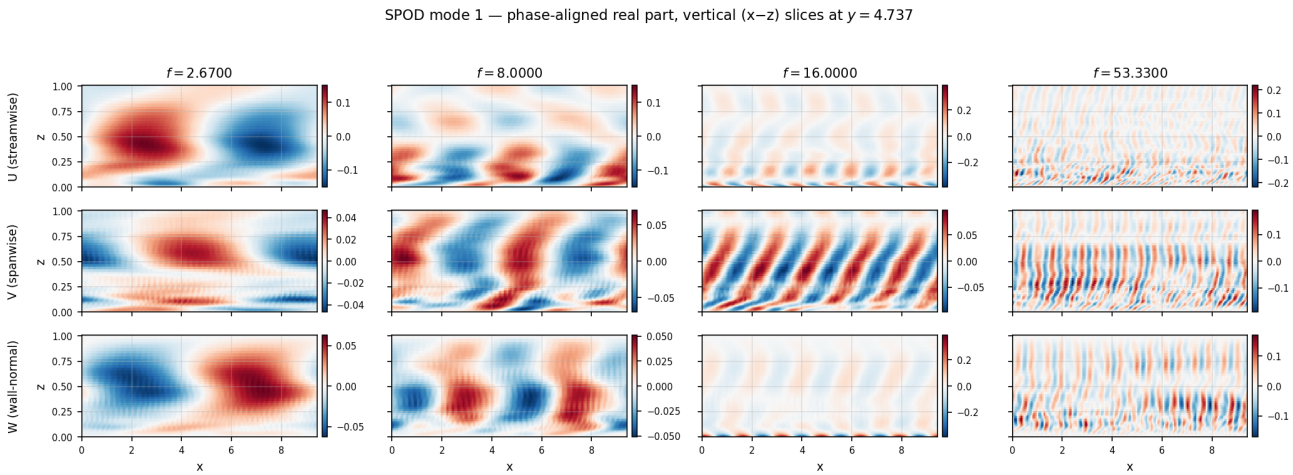

Figure 3: Phase-aligned real part of SPOD mode 1 at four frequencies ($f = 2.67$, 8.0, 16.0, 53.33), plotted as vertical (x–z) slices at mid-span ($y = 4.737$). Rows: U, V, W velocity components. Columns: increasing frequency from left to right. Each panel is normalised to its own maximum amplitude.

Mode 1: dominant coherent structure

The frequency progression of mode 1 (Figure 3) reveals a systematic transformation in spatial character. At $f = 2.67$ the structure is large and smooth, spanning the full streamwise extent with strong streamwise dominance ($U : V : W = 68.0\% : 21.6\% : 10.4\%$) and energy centred in the lower-to-mid domain ($z_{\text{peak}} = 0.39$, $\bar{z} = 0.49$). By $f = 8.0$ the structure contracts and develops a forward-leaning inclination consistent with mean-shear deformation. At $f = 16.0$ the behaviour changes qualitatively: W rises to 35.5%, approaching parity with U (44.1%) — the only frequency in the spectrum at which this occurs — and the mode collapses onto a thin near-surface layer. At $f = 53.33$ all structure is lost and the partition approaches near-isotropy ($U : V : W \approx 26\% : 41\% : 33\%$).

Modes 2 and 3: independent sub-dominant structures

At $f = 2.67$, modes 1, 2, and 3 collectively account for approximately 65% of the total spectral energy. The sub-dominant modes differ from mode 1 in both wall-normal localisation and component partition: mode 2 is concentrated in the upper half of the domain ($z \gtrsim 0.5$), while mode 3 is confined to the lower quarter ($z \lesssim 0.25$). Their component partitions ($U : V : W = 72.2\% : 17.7\% : 10.1\%$ and $76.0\% : 14.6\% : 9.5\%$ respectively) are

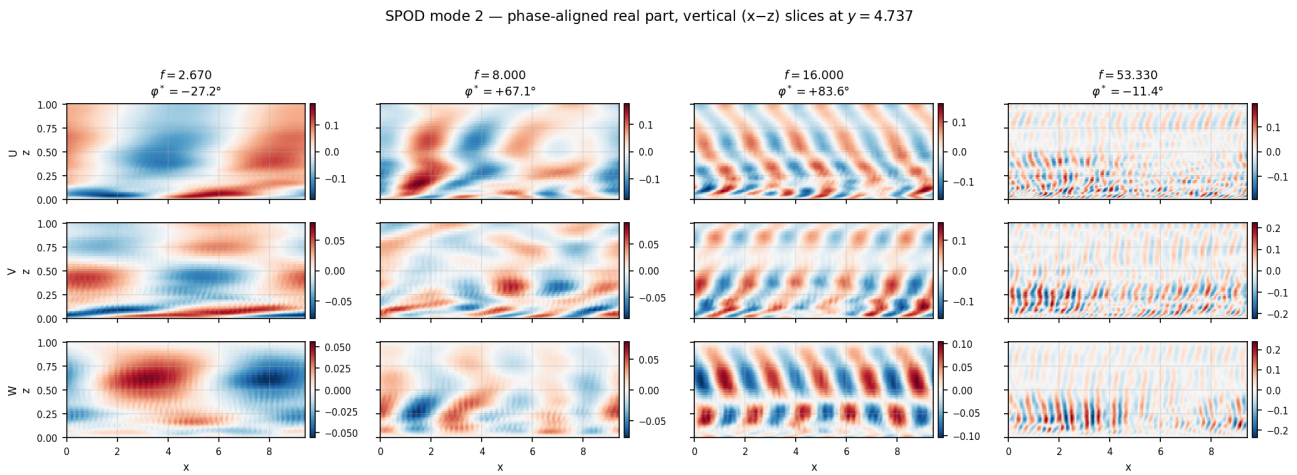

Figure 4: Phase-aligned real part of SPOD mode 2 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 3.

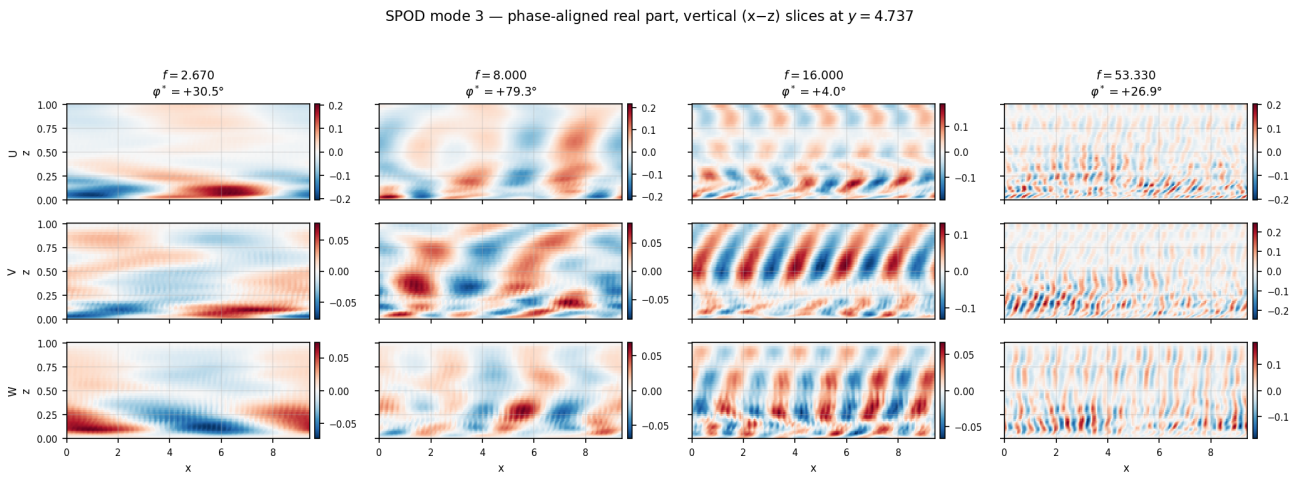

Figure 5: Phase-aligned real part of SPOD mode 3 at four frequencies, plotted as vertical (x–z) slices at mid-span. Layout identical to Figure 3.

systematically more U-dominated than mode 1, and their spatial wavelengths differ. These differences in wall-normal localisation and streamwise scale rule out the harmonic and phase-shifted-copy hypotheses, identifying the three modes as distinct coexisting structures. At $f = 53.33$ all three modes converge to a near-identical incoherent character ($U : V : W \approx 26$–$27\% : 40$–$41\% : 33\%$).

Component-energy cascade across rank and frequency

A consistent quantitative trend emerges across the three modes. Within the coherent regime, the streamwise share U *increases* with mode rank ($68.0\% \to 72.2\% \to 76.0\%$ at $f = 2.67$), reflecting an increasingly streak-dominated character in higher-rank modes. With increasing frequency, the U share *decreases* for every individual mode and the partition approaches isotropy. By $f = 53.33$, all three modes share the same mildly V-biased near-isotropic split. The streamwise-anisotropy imprint of the wave-driven shear is therefore strongest in the leading mode at the lowest coherent frequency and is progressively erased moving toward either higher rank or higher frequency.

3.4 Wall-Normal Energy Distribution and Component-Energy Cascade

Two quantitative diagnostics were extracted from mode 1 to complement the spatial visualisation of Section 3.3. Panel (a) of Figure 6 presents the wall-normal energy distribution $E(z)$ at the four target frequencies, and panel (b) presents the U, V, W component-energy shares across all 26 resolved frequencies.

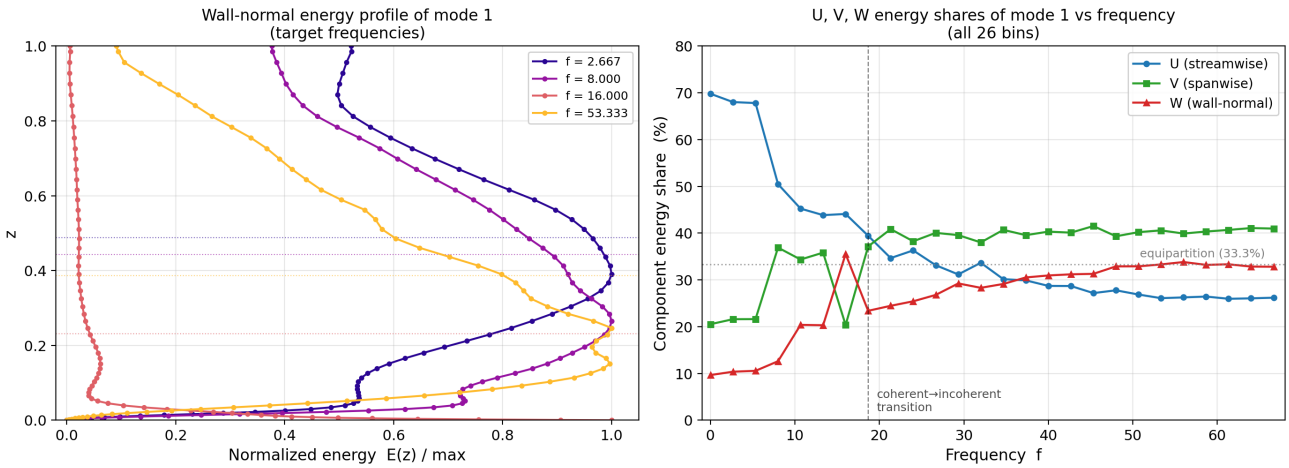

Figure 6: (a) Wall-normal energy profile of SPOD mode 1 at the four target frequencies, normalised to the profile peak. Dotted horizontal lines mark the energy centroids $\bar{z}$. (b) U, V, and W component-energy shares of mode 1 across all 26 frequency bins. The grey dotted line marks equipartition (1/3) and the vertical dashed line marks the coherent–incoherent transition.

Wall-normal energy profiles

The four profiles are qualitatively distinct, confirming that each target frequency samples a different wall-normal organisation. The key diagnostics are summarised in Table 2.

Table 2: Wall-normal localisation of SPOD mode 1 at the four target frequencies. z_{peak} is the height of the energy maximum and $\bar{z}$ is the volume-weighted centroid. The integral $\int E(z)\,\mathrm{d}z = 1$ in all cases confirms mode orthonormality.

f	z_{peak}	$\bar{z}$	$\int E(z)\,\mathrm{d}z$
2.67	0.390	0.488	1.000
8.0	0.265	0.443	1.000
16.0	0.000	0.231	1.000
53.33	0.247	0.386	1.000

A clear progression emerges: the primary coherent mode ($f = 2.67$) peaks at $z_{\text{peak}} = 0.39$ in the lower-to-mid domain ($\bar{z} = 0.49$), and the peak height decreases monotonically with increasing frequency through the intermediate coherent range (f = 8.0, 13.33). At $f = 16.0$, the mode becomes uniquely surface-locked ($z_{\text{peak}} = 0.000$, $\bar{z} = 0.23$) — the only frequency at which all energy concentrates at the domain boundary. The incoherent reference ($f = 53.33$) shows a bimodal profile with a secondary peak near $z = 0.95$ and primary energy near $z = 0.25$, reflecting the breakdown of coherence into spatially uncorrelated fluctuations.

Component-energy cascade across frequency

Panel (b) reveals a systematic anisotropy cascade in the leading mode. In the primary coherent regime ($f \lesssim 5.33$), the mode is strongly U-dominated ($U : V : W \approx 70 : 21 : 10$). Across the intermediate coherent range, the U share drops sharply while W peaks at 35.5% at $f = 16.0$ — the only frequency at which W exceeds V. Beyond the coherent–incoherent transition at $f \approx 18.67$, all three shares stabilise: U crosses equipartition at $f = 26.67$ and the partition settles at $U : V : W \approx 27 : 40 : 33$, indicating near-isotropic but mildly V-biased small-scale fluctuations.

4 How the code works?

This section bridges the theoretical formulation of Section 2 with a memory-efficient discrete implementation. The pipeline is structured around four phases: data ingestion and matrix setup, preprocessing (windowing and fluctuation extraction), spectral estimation, and mode reconstruction.

4.1 Data Ingestion and Matrix Setup

The raw DNS data, originally generated by Fortran using unformatted stream access (`ACCESS='STREAM'`), exists as a continuous sequence of binary bytes without record markers. Because the SPOD algorithm requires rapid,

non-sequential access to overlapping time segments, the data must be structurally rearranged. The algorithm systematically reads the raw stream block by block, identifying exactly $N_x \times N_y \times N_z$ scalar values for each of the three velocity components per timestep. These sequential blocks are concatenated to build the structured 2D time-domain snapshot matrix Q, where rows represent flattened spatial coordinates and columns represent discrete time snapshots. To manage the data footprint and allow rapid access to time blocks, the rearranged matrix Q is written directly to disk as a memory-mapped array via the `build_Q_memmap` routine, circumventing system memory limitations during processing.

4.2 Spatial Weight Matrix Computation

The discrete weight matrix W defines the inner product under which the SPOD eigenvalue problem (1) is posed. Physically, W acts as a numerical quadrature rule that accounts for the varying spatial measure of each grid point.

For the present computational domain, the horizontal grid spacing is uniform ($\Delta x = L_x/N_x$ and $\Delta y = L_y/N_y$), while the vertical coordinates z_k are strictly increasing but non-uniform. The discrete vertical weight at index k is computed using the trapezoidal rule:

$$\Delta z_k = \tfrac{1}{2}(z_{k+1} - z_{k-1}), \tag{11}$$

with one-sided differences applied at the domain boundaries. The local spatial weight for any node is the three-dimensional control volume $\Delta V_{ijk} = \Delta x\, \Delta y\, \Delta z_k$.

Algorithmic consistency is verified by ensuring that the sum of all nodal weights exactly reproduces the physical domain volume:

$$\sum_{k=0}^{N_z-1} \sum_{j=0}^{N_y-1} \sum_{i=0}^{N_x-1} \Delta x\, \Delta y\, \Delta z_k = L_x L_y\, (z_{\max} - z_{\min}). \tag{12}$$

For the present grid, the computed weight sum matches the analytical value $9\pi^2 \approx 88.83$ to machine precision.

Because the state vector consists of three dimensionally identical velocity components, no specialised variable-scaling is required. The full weight matrix W is constructed as a 1D diagonal array by tiling the localised grid volumes across the three velocity components, ensuring that every component is scaled identically by the physical mesh geometry.

4.3 Block Segmentation and Fluctuation Extraction

To estimate a robust CSD, the snapshot matrix Q is partitioned into an ensemble of shorter temporal blocks. If the blocks were extracted sequentially without overlap (as in the Bartlett method[5]), the abrupt truncation at the block boundaries would introduce artificial high-frequency noise into the Fourier transform — spectral leakage. To mitigate this, a Hann window[6] is applied to each block, smoothly tapering the velocity signal to zero at the beginning and end of the segment and enforcing strict periodicity.

The Hann window introduces a secondary complication: by forcing the boundaries to zero, it discards physical content near the edges of every block. To compensate, the algorithm uses Welch's method with 50% block overlap, ensuring that turbulence attenuated at the edge of one block sits at the unattenuated peak of an adjacent one. The overlap also increases the number of available block realisations, raising the statistical degrees of freedom and reducing the variance of the spectral estimate. Because true turbulent structures are coherent across time, ensemble-averaging across many overlapping blocks suppresses random noise while preserving distinct energy peaks of the underlying physical phenomena.

The matrix is loaded into memory piece by piece by selectively extracting block-specific columns from the memmap. Once a block is loaded, the temporal mean of that specific block is calculated and subtracted from every row, applying a Reynolds decomposition to convert raw velocity data into purely turbulent fluctuation data:

$$q'(\mathbf{x}, t) = q(\mathbf{x}, t) - \bar{q}_{\text{block}}(\mathbf{x}). \tag{13}$$

Performing the mean subtraction at this stage — rather than having the Fortran solver export pre-computed fluctuations — is a deliberate architectural choice. Configuring the DNS to export the full instantaneous velocity field preserves the complete physical state of the flow and maximises the dataset's flexibility, allowing the same binary output to be reused for different analyses without re-running the simulation. The SPOD pipeline therefore takes responsibility for isolating the fluctuations.

From a fluid-dynamics perspective, SPOD is designed to extract dynamic, coherent turbulent structures interacting at specific non-zero frequencies. If the stationary mean flow $\bar{q}$ were retained, its kinetic energy would dominate the decomposition and manifest as an overwhelming zero-frequency mode. By subtracting the block mean, the algorithm isolates fluctuations relative to the instantaneous baseline velocity profile and directs the decomposition toward transient eddy structures at non-zero frequencies. The residual energy visible at $f = 0$ reflects block-to-block variations in the mean flow that persist despite this preprocessing — a feature expected when the baseline state is not strictly stationary.

4.4 Spectral Transformation and Frequency Matrix Assembly

Following preprocessing, the algorithm transforms the data into the frequency domain. For every overlapping block, a discrete real FFT is applied along the temporal dimension. To preserve physical energy, the resulting Fourier coefficients are scaled by the factor

$$\kappa = \sqrt{\frac{\Delta t}{\sum_{n=1}^{N_{\text{blk}}} w_n^2}}, \tag{14}$$

where w_n is the amplitude of the Hann window at sample n and Δt is the snapshot time spacing. This scaling compensates for the variance attenuation introduced by the windowing and ensures that the final eigenvalues $\hat{\Lambda}(f)$ represent power spectral density with correct physical dimensions.

In classical SPOD theory, applying the FFT to all blocks yields a three-dimensional tensor of size $N_{\text{space}} \times N_{\text{blk}} \times N_f$. Holding this tensor in memory is computationally prohibitive for high-fidelity DNS datasets. The present implementation evaluates the decomposition one frequency at a time: for a targeted discrete frequency f_j, the algorithm extracts the corresponding complex Fourier coefficients from every block and concatenates them into the localised 2D frequency-domain snapshot matrix $X(f_j) \in \mathbb{C}^{N_{\text{space}} \times N_{\text{blk}}}$. This isolated matrix contains all the spatial and statistical information required for the reduced eigenvalue problem at the chosen frequency.

4.5 Assembly of the Reduced Correlation Matrix

The pipeline is structured to solve the reduced eigenvalue problem of Equation (5). While the theoretical construction of R assumes a monolithic matrix product X^*WX, computing this simultaneously over the full spatial domain exceeds standard memory limits.

To circumvent this, the algorithm exploits the distributive property of matrix multiplication. The spatial domain is divided into manageable segments (chunks of 20,000 grid points). For each spatial chunk, the algorithm extracts the corresponding rows of $X(f_j)$, multiplies them by their localised spatial weights from W, and computes the partial inner product. These partial matrices are cumulatively summed across all spatial chunks and finally ensemble-averaged by the block count N_{blk}. This chunking strategy yields the exact R matrix without ever loading the full $X(f_j)$ into memory simultaneously.

While R is theoretically guaranteed to be Hermitian, the chunked floating-point operations introduce microscopic numerical roundoff. To ensure strict numerical stability for the eigenvalue solver, Hermitian symmetry is explicitly enforced on the final matrix by averaging it with its conjugate transpose, $R \leftarrow \frac{1}{2}(R + R^*)$, before decomposition.

4.6 Reduced Eigenvalue Decomposition

Once $R(f_j)$ is constructed and stabilised, the algorithm solves the Hermitian eigenvalue problem of Equation (5). An optimised linear-algebra solver extracts the discrete modal energies $\hat{\Lambda}$ and the corresponding orthonormal reduced eigenvectors Ψ. Because SPOD's primary objective is to isolate the most dominant coherent flow structures, the resulting eigenvalues (and associated eigenvectors) are sorted in strictly descending order, so that the first mode $\hat{\Phi}_1$ is mathematically guaranteed to represent the most energetic phenomenon at the targeted frequency f_j.

A numerical-tolerance filter is applied to the eigenvalues. While true physical kinetic energy must be strictly non-negative, the floating-point operations involved in building R can produce microscopic non-physical negative eigenvalues at the level of machine epsilon ($\sim 10^{-16}$). The algorithm identifies these numerical artifacts and forces them to zero, ensuring physical validity before the spatial reconstruction phase.

4.7 Spatial Mode Reconstruction and Verification

With $\hat{\Lambda}$ and Ψ extracted, the pipeline reconstructs the 3D spatial modes $\hat{\Phi}$ via Equation (9). Just as for the correlation matrix assembly, computing $\hat{\Phi} = (N_{\text{blk}}\hat{\Lambda})^{-1/2} X\Psi$ directly exceeds available memory. The pipeline therefore reads the frequency matrix $X(f_j)$ in spatial segments, projects each segment onto the scaled eigenvectors, and streams the resulting spatial modes directly to disk.

As a final validation step, the orthonormality of the reconstructed modes is verified by a localised, chunked computation of $\hat{\Phi}^*W\hat{\Phi}$ and comparison to the identity matrix. The maximum residual error of $\sim 10^{-15}$ confirms that the extracted modes are physically independent, orthogonal, and correctly scaled with respect to the non-uniform grid volume.

4.8 Eigenvalues as Power Spectral Density

The eigenvalues $\hat{\Lambda}_k(f_j)$ computed at each frequency represent the contribution of the k-th spatial mode to the kinetic-energy spectrum at that frequency. Because the Fourier coefficients were scaled by κ (Equation (14)), the eigenvalues inherit the physical dimension of power spectral density: energy per unit frequency.

The sum $\sum_{k=1}^{N_{\mathrm{blk}}} \hat{\Lambda}_k(f_j)$ yields the total kinetic energy density at frequency f_j. Integrating across all frequencies recovers the total turbulent kinetic energy of the flow:

$$\mathrm{TKE} = \sum_{f_j} \sum_{k=1}^{N_{\mathrm{blk}}} \hat{\Lambda}_k(f_j)\, \Delta f, \tag{15}$$

where Δf is the frequency resolution. The SPOD spectrum therefore quantifies the distribution of kinetic energy across both spatial scales (mode index k) and temporal scales (frequency f), and the dominant eigenvalue at the peak frequency identifies both the location of the strongest coherent structure in the spectrum and its energetic importance relative to the turbulent cascade.

4.9 Full Spectrum Execution

With the single-frequency pipeline validated, the algorithm iterates over the full set of discrete frequencies returned by the real FFT. The physical frequency axis is determined by the snapshot sampling rate Δt.

To ensure stable execution without exceeding memory limits or encountering operating-system file-lock conflicts, the pipeline explicitly purges memory-mapped file handles and triggers Python garbage collection at the start of each frequency iteration. The spatial orthonormality verification ($\hat{\Phi}^* W \hat{\Phi} = I$) is restricted to the initial frequency step to minimise computational overhead, since the projection's mathematical structure is invariant across frequencies. As the pipeline iterates through the spectrum, the localised 3D spatial modes $\hat{\Phi}$ are written directly to disk while the modal energies $\hat{\Lambda}$ are aggregated into a global data structure. This aggregation yields the complete SPOD energy spectrum, which serves as the primary diagnostic tool for identifying coherent turbulent structures and their characteristic frequencies.

A Complete SPOD Eigenvalue Spectrum

Table 3: Full SPOD spectral data across all 26 resolved frequencies ($N_{\text{blk}} = 19$ blocks). Total summed modal energy: $\sum_f \sum_k \hat{\Lambda}_k(f) = 1.512 \times 10^{-1}$.

idx	f	$\sum \hat{\Lambda}_k$	$\hat{\Lambda}_1$	$\hat{\Lambda}_1 / \sum \hat{\Lambda}_k$	$\hat{\Lambda}_1 / \hat{\Lambda}_2$	**Modes** $\geq 90\%$
0	0.00	1.2409e-02	2.6916e-03	0.217	1.21	8
1	2.67	2.6058e-02	8.8353e-03	0.339	1.97	10
2	5.33	2.3090e-02	3.4933e-03	0.151	1.14	14
3	8.00	1.6923e-02	1.9583e-03	0.116	1.19	16
4	10.67	1.2894e-02	1.9019e-03	0.148	1.53	16
5	13.33	9.8843e-03	1.0629e-03	0.108	1.20	16
6	16.00	8.6232e-03	1.5526e-03	0.180	2.09	16
7	18.67	6.5599e-03	5.3657e-04	0.082	1.05	16
8	21.33	5.1826e-03	4.0844e-04	0.079	1.05	17
9	24.00	4.3324e-03	3.4575e-04	0.080	1.10	17
10	26.67	3.6861e-03	3.0333e-04	0.082	1.05	17
11	29.33	3.1061e-03	2.7042e-04	0.087	1.14	17
12	32.00	2.6392e-03	2.0960e-04	0.079	1.04	17
13	34.67	2.2513e-03	1.7357e-04	0.077	1.06	17
14	37.33	1.9313e-03	1.4564e-04	0.075	1.05	17
15	40.00	1.6804e-03	1.3238e-04	0.079	1.07	17
16	42.67	1.4699e-03	1.0954e-04	0.075	1.03	17
17	45.33	1.2952e-03	9.6010e-05	0.074	1.05	17
18	48.00	1.1652e-03	8.5129e-05	0.073	1.03	17
19	50.67	1.0495e-03	7.7515e-05	0.074	1.05	17
20	53.33	9.5332e-04	7.0977e-05	0.074	1.05	17
21	56.00	8.7816e-04	6.3759e-05	0.073	1.04	17
22	58.67	8.2588e-04	5.9304e-05	0.072	1.02	17
23	61.33	7.8814e-04	5.6850e-05	0.072	1.01	17
24	64.00	7.6450e-04	5.4697e-05	0.072	1.01	17
25	66.67	7.5853e-04	5.4538e-05	0.072	1.02	17
Total	—	1.5120e-01	—	—	—	—

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